

Sustainable Development

SUSTAINABLE DEVELOPMENT

Sustainable Development and Environmental issues:

Sustainable development: Sustainable development is *development that meets the needs of the present without compromising the ability of future generations* to meet their own needs. It contains within it two key concepts:

- the concept of '**needs**', in particular the essential needs of the world's poor, to which overriding priority should be given; and
- the idea of **limitations** imposed by the state of technology and social organization on the environment's ability to meet present and future needs.

For sustainable development to be achieved, it is crucial to harmonize three core elements: **economic growth, social inclusion and environmental protection**. These elements are interconnected and all are crucial for the well-being of individuals and societies.

Evolution of Sustainable Development

Brundtland Commission

The above definition of Sustainable Development was given by **Brundtland Commission**. Brundtland Commission was a commission established by the **28 United Nations in 1983 as World Commission on Environment and Development (WCED)**. The commission was created to address the growing concern **"about the accelerating deterioration of the human environment and natural resources and the consequences of that deterioration for economic and social development."**

Thereafter, in 1987, the World Commission on Environment and Development submitted its report, which is also known as **Brundtland Commission Report** wherein an effort was made to link economic development and environment protection.

In **1992, Rio Declaration on Environment and Development codified the principle of Sustainable Development**.

The outcome of this commission was the "Brundtland Report". The title of this report was **"Our Common Future"**. This report gave the definition: ***"Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."*** and this definition is quoted first of all when anybody discusses about the sustainability.

Our Common Future is also known as the **Brundtland Report** in recognition of former Norwegian Prime Minister Gro Harlem Brundtland's role as *Chair of the World Commission on Environment and Development.

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Earth Charter/Earth Summit

“Our Common Future” the report of the **“Brundtland Commission”** came out with a new guide to sustainable development. The “Brundtland Commission”, called for “a universal declaration” and “new charter” to set “new norms” to guide the transition to sustainable development.

The Earth Charter was proposed during the preparatory process to the UN Conference on Environment and Development — best known as the Earth Summit — held in Rio de Janeiro, Brazil, in 1992.

After the Rio Summit or Earth Summit in 1992, **in 1994, Maurice Strong (Chairman of the Earth Summit)** and Mikhail Gorbachev, working through organizations they each founded (the Earth Council and Green Cross International respectively), restarted the Earth Charter as a civil society initiative, with the help of the government of the Netherlands. Earth Charter is a 2,400 word document divided into 4 sections, called four pillars and sixteen main principles containing sixty-one supporting principles.

Millennium Development Goals

Between **2000 and 2015**, there were **8 Millennium Development Goals** with 21 targets. The MDGs targeted developing countries, particularly the poorest, while the SDGs apply to all countries, developed and developing.

SUSTAINABLE DEVELOPMENT GOALS (SDGS)

The 193 Member States of the United Nations officially adopted a new sustainable development agenda entitled, **“Transforming Our World: The 2030 Agenda for Sustainable Development”** at the Sustainable Development Summit held at UN Headquarters in New York in September 2015. This agenda contains **17 Goals and 169 targets**.

The Sustainable Development Goals (SDGs) **start on 1 January 2016 and are expected to be achieved by 31 December 2030**. However, some targets that build on pre-set international agreements are expected to be achieved sooner.

This universal, integrated and transformative agenda aims to spur actions that will end poverty and build a more sustainable world over the next 15 years. This agenda builds on the achievements of the Millennium Development Goals (MDGs), which were adopted in 2000 and guided development action for the last 15 years. The MDGs have proven that global goals can lift millions out of poverty.

The decision to launch a process to develop a set of SDGs was made by UN Member States at the United Nations Conference on Sustainable Development (Rio+20), held in Rio de Janeiro in June 2012.

The **Addis Ababa Action Agenda** that came out of the **Third International Conference on Financing for Development held in Addis Ababa from 13-16 July 2015**, provided concrete policies and actions to support the **implementation of the new agenda for SDGs**.

At the global level, the **17 Sustainable Development Goals (SDGs) and 169 targets** of the new agenda will be monitored and reviewed using a set of global indicators. The global indicator

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framework, to be developed by the Inter Agency and Expert Group on SDG Indicators (IAEA-SDGs).

The Goals and targets will stimulate action over the next 15 years in areas of critical importance: **people, planet, prosperity, peace and partnership.**

- **People** - to end **poverty and hunger**, in all their forms and dimensions, and to ensure that all human beings can fulfil their potential in **dignity and equality** and in a healthy environment.
- **Planet** - to protect the planet from degradation, including through sustainable consumption and production, sustainably managing its natural resources and taking urgent action on climate change, so that it can support the needs of the present and future generations.
- **Prosperity** - to ensure that all human beings can enjoy prosperous and fulfilling lives and that economic, social and technological progress occurs in harmony with nature.
- **Peace** - to foster peaceful, **just and inclusive societies free from fear and violence.** There can be no sustainable development without peace and no peace without sustainable development.
- **Partnership** - to mobilize the means required to implement this agenda through a revitalized global partnership for sustainable development, based on a spirit of strengthened global solidarity, focused in particular on the needs of the poorest and most vulnerable and with the participation of all countries, all stakeholders and all people.

List of Sustainable Development Goals

- **Goal 1. End poverty** in all its forms everywhere
- **Goal 2. End hunger**, achieve food security and improved nutrition and promote sustainable agriculture

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- **Goal 3. Ensure healthy** lives and promote well-being for all at all ages
- **Goal 4. Ensure inclusive and equitable quality education** and promote lifelong learning opportunities for all
- **Goal 5. Achieve gender equality** and empower all women and girls
- **Goal 6.** Ensure availability and sustainable management of **water and sanitation for all**
- **Goal 7.** Ensure **access to affordable, reliable, sustainable and modern energy for all**
- **Goal 8.** Promote **sustained, inclusive and sustainable economic growth**, full and productive employment and decent work for all
- **Goal 9.** Build **resilient infrastructure**, promote inclusive and sustainable industrialization and foster innovation
- **Goal 10. Reduce inequality** within and among countries
- **Goal 11.** Make **cities and human settlements inclusive, safe, resilient** and sustainable
- **Goal 12.** Ensure **sustainable consumption and production** patterns
- **Goal 13.** Take urgent action to **combat climate change and its impacts***
- **Goal 14. Conserve and sustainably use the oceans, seas and marine resources** for sustainable development
- **Goal 15.** Protect, restore and promote **sustainable use of terrestrial ecosystems**, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
- **Goal 16.** Promote **peaceful and inclusive societies for sustainable development**, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
- **Goal 17.** Strengthen the means of **implementation and revitalize the global partnership** for sustainable development

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Mnemonics to read SDG:



I HOPE We Can

- I** → Innovation (Goal 9)
Inequality (Goal 10)
- H** → Hunger (Goal 2)
Health (Goal 3)
- O** → Ocean (Goal 14)
- P** → Poverty (Goal 1)
Peace (Goal 16)
Partnership (Goal 17)
- E** → Education (Goal 4)
Energy (Goal 7)
Economic Growth (Goal 8)
Ecosystem (Goal 15)
- W** → Women (Goal 5)
Water (Goal 6)
- C** → Cities (Goal 11)
Consumption (Goal 12)
Climate Action (Goal 13)

By Letter Count:

1. Women → Goal 5 (5 Letters)
2. Economic → Goal 8 (8 Letters)
3. Inequality → Goal 10 (10 Letters)
4. Healthy City → Goal 11 (11 Letters)
5. Climate Change → Goal 13 (13 Letters)
6. Life below water → Goal 14 (14 Letters)
7. Global partnership → Goal 17 (17 Letters)

By the position of Letter:

1. I → Innovation: Goal 9 (9th Letter)
2. P → Peace: Goal 16 (16th Letter)

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CLIMATE CHANGE:

United Nations Framework Convention on Climate Change (UNFCCC):

It was adopted at the **Earth Summit in Rio de Janeiro, Brazil, in June 1992**. The Convention entered into force in 1994 and **now has 198 parties**.

Its original objective was the **stabilization of GHG concentrations in the atmosphere by 2000** at a level that would prevent dangerous anthropogenic interference in the climate system, but it did not call for parties to reduce emissions of carbon dioxide, methane and other GHGs that have contributed to a warming of the Earth's mean surface temperature.

The **United Nations Climate Change Conferences** are **yearly conferences** held in the **framework of the United Nations Framework Convention on Climate Change (UNFCCC)**. They serve as the formal meeting of the UNFCCC Parties (**Conference of the Parties, COP**) to assess progress in dealing with climate change, and beginning in the mid-1990s, to negotiate the Kyoto Protocol to establish legally binding obligations for developed countries to reduce their greenhouse gas emissions.

From 2005 the Conferences have also served as the "Conference of the Parties Serving as the Meeting of Parties to the Kyoto Protocol" (CMP).

The secretariat of the United Nations Framework Convention on Climate Change is located in Bonn, Germany.

KYOTO PROTOCOL:

The Kyoto Protocol (COP3) is an **international agreement linked to the United Nations Framework Convention on Climate Change**, which commits its Parties by **setting internationally binding emission reduction targets**.

Recognizing that developed countries are principally responsible for the current high levels of GHG emissions in the atmosphere as a result of more than 150 years of industrial activity, the Protocol places a heavier burden on developed nations under the principle of "**common but differentiated responsibilities**."

The Kyoto Protocol was adopted in Kyoto, Japan, **on 11 December 1997** and entered into force on **16 February 2005**.

The detailed rules for the implementation of the Protocol were **adopted at COP 7** in Marrakesh, Morocco, in 2001, and are referred to as the "**Marrakesh Accords**." Its first commitment period started in 2008 and ended in 2012. The Kyoto Protocol has two commitment periods: 2008-12 and 2013-20.

By 2008-2012, countries have to reduce their GHG **emissions by an average of 5% below their 1990 levels**

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The second commitment period was agreed on in 2012, known as **Doha Amendment to the Protocol**. **Parties committed to reducing GHG** emissions by at **least 18% below 1990** levels in the eight-year period from 2013 to 2020 during the second commitment period;

The Kyoto Protocol is an international treaty which extends the 1992 United Nations Framework Convention on Climate Change (UNFCCC) that commits State Parties to reduce greenhouse gas emissions, based on the scientific consensus that

- a) global warming is occurring and
- b) it is extremely likely that human-made CO₂ emissions have predominantly caused it..

The Kyoto Protocol emission target gases include

1. **Carbon dioxide (CO₂),**
2. **Methane (CH₄),**
3. **Nitrous oxide (N₂O),**
4. **Sulphur hexafluoride (SF₆),**
5. **groups of hydro fluorocarbons (HCFs) and**
6. **groups of Per fluorocarbons (PFCs).**

The Kyoto mechanisms are:

- **International Emissions Trading:** Parties with commitments under the Kyoto Protocol **have accepted different targets for limiting or reducing emissions**. These targets are expressed as **levels of allowed emissions, or “assigned amounts,” over the 2008-2012** commitment period. The allowed emissions are divided into “assigned amount units” (**AAUs**). Emissions trading, as set out in Kyoto Protocol, allows countries that have emission units to spare - emissions permitted them but not "used" - to sell this excess capacity to countries that are over their targets. Thus, a **new commodity was created in the form of emission reductions or removals**. Since carbon dioxide is the principal greenhouse gas, people speak simply of trading in carbon. Carbon is now tracked and traded like any other commodity. This is known as the **"carbon market."**

Clean Development Mechanism (CDM):

- The **Clean Development Mechanism (CDM)**, defined in **Article 12 of the Protocol**, allows a country with an **emission-reduction or** emission-limitation commitment under the Kyoto Protocol to implement an emission-reduction project in developing countries. Such projects can earn saleable certified emission reduction (CER) credits, each equivalent to one tonne of CO₂, which can be counted towards meeting Kyoto targets.
- E.g. of CDM: USA takes up or finances some environment benefitting project in India (solar power projects, wind power projects, afforestation etc.) and earns some **carbon credits (certified emission reduction credits)**. Now it shows these earned carbon credits to the world and tells them how it is working towards meeting its Kyoto targets. Such projects can earn saleable certified emission reduction (CER) credits, each equivalent to one tonne of CO₂, which can be counted towards meeting Kyoto targets.

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- The CDM is the main source of income for the UNFCCC Adaptation Fund, which was established to finance adaptation projects and programmes in developing country Parties to the Kyoto Protocol that are particularly vulnerable to the adverse effects of climate change. The Adaptation Fund is financed by a 2% levy on Certified Emissions Reduction issued by the CDM.
- **Joint implementation (JI):** The mechanism known as "**joint implementation**", defined in Article 6 of the Kyoto Protocol, allows a country with an emission reduction or limitation commitment under the Kyoto Protocol to earn emission reduction units (ERUs) from an emission-reduction or emission removal project in another country, each equivalent to one tonne of CO₂, which can be counted towards meeting its Kyoto target. As the host Party gains from foreign investment and knowledge transfer, the joint implementation gives Parties a flexible and economical way to fulfill a portion of their Kyoto commitments.

Targets for the first commitment period: The targets for the first commitment period of the Kyoto Protocol cover emissions of the six main greenhouse gases, namely.

- Carbon dioxide (CO₂);
- Methane (CH₄);
- Nitrous oxide (N₂O);
- Hydrofluorocarbons (HFCs);
- Perfluorocarbons (PFCs); and
- Sulphur hexafluoride (SF₆)

ADAPTATION FUND (AF):

It was established in **2001** to finance concrete adaptation projects and programmes in **developing country** that are **Parties to the Kyoto Protocol** of the United Nations Framework Convention on Climate Change (UNFCCC) that are particularly vulnerable to the adverse effects of climate change.

- **Financing:** The Adaptation Fund is financed with a share of proceeds from the clean development mechanism (CDM) project activities and other sources of funding. The share of proceeds amounts to **2 per cent of certified emission reductions (CERs)** issued for a CDM project activity.
- **Administrative Management:** The Adaptation Fund is supervised and managed by the **Adaptation Fund Board (AFB)**. The AFB is composed of 16 members and 16 alternates and meets at least twice a year (Membership of the AFB).
 - The **World Bank (WB)** serves as trustee of the Adaptation Fund on an interim basis. **WB Secretariat: Washington DC**

Since 2010, the Adaptation Fund has committed over 1 billion for climate change adaptation and resilience projects and programmes, including 160 concrete, localized projects in the most vulnerable communities of developing countries around the world with over 43 million total beneficiaries.

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PARIS AGREEMENT:

The Paris Agreement is a **legally binding** international treaty on climate change. It was adopted by 196 Parties at the **UN Climate Change Conference (COP21) in Paris, France, on 12 December 2015**. It entered into force on 4 November 2016.

The Paris Agreement opened for signature on 22 April 2016 – Earth Day – at UN Headquarters in New York. **It entered into force on 4 November 2016, 30 days after the so-called “double threshold” (ratification by 55 countries that account for at least 55% of global emissions)** had been met.

Some of the key aspects of the Agreement are set out below:

1. Long-term temperature goal – The Paris Agreement, in seeking to strengthen the global response to climate change, reaffirms that its overarching goal is to hold “the increase in the global average temperature to **well below 2°C above pre-industrial levels**” and pursue efforts “to limit the temperature increase **to 1.5°C above pre-industrial levels**.”

2. Global peaking and 'climate neutrality' – To achieve this temperature goal, Parties aim to reach global peaking of greenhouse gas emissions (GHGs) as soon as possible, recognizing peaking will take longer for developing country Parties, so as to achieve a balance between anthropogenic emissions by sources and removals **by sinks of GHGs in the second half of the century**.

However, in recent years, world leaders have stressed the need to limit global warming to 1.5°C by the end of this century. That’s because the UN’s Intergovernmental Panel on Climate Change indicates that crossing the 1.5°C threshold risks unleashing far more severe climate change impacts, including more frequent and severe droughts, heatwaves and rainfall.

To limit global warming to 1.5°C, **greenhouse gas emissions must peak before 2025 at the latest and decline 43% by 2030.**

The Paris Agreement is **a landmark** in the multilateral climate change process because, for the first time, a binding agreement brings all nations together to combat climate change and adapt to its effects.

3. Mitigation – The Paris Agreement establishes binding commitments by all Parties to prepare, communicate and maintain a **nationally determined contribution (NDC)** and to pursue domestic measures to achieve them. It also prescribes that Parties shall **communicate their NDCs every 5 years** and provide information necessary for clarity and transparency.

4. Sinks and reservoirs – The Paris Agreement also encourages Parties to conserve and enhance, as appropriate, sinks and reservoirs of GHGs.

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5. Voluntary cooperation/Market- and non-market-based approaches – The Paris Agreement recognizes **the possibility of voluntary cooperation** among Parties to allow for higher ambition and sets out principles – including **environmental integrity, transparency and robust accounting** – for any cooperation that involves international transfer of mitigation outcomes

6. Adaptation – The Paris Agreement establishes a **global goal on adaptation** – of enhancing adaptive capacity, strengthening resilience and reducing vulnerability to climate change in the context of the temperature goal of the Agreement. It aims to significantly strengthen national adaptation efforts, including through support and international cooperation. It recognizes that adaptation is a global challenge faced by all

7. Loss and damage – The Paris Agreement recognizes the importance of **averting, minimizing and addressing loss and damage associated** with the adverse effects of climate change, including extreme weather events and slow onset events, and the role of sustainable development in reducing the risk of loss and damage.

8. Finance, technology and capacity-building support– The Paris Agreement reaffirms the obligations of developed countries **to support the efforts of developing country Parties to build clean, climate-resilient futures**, while for the first time encouraging voluntary contributions by other Parties. The agreement also provides that the Financial Mechanism of the Convention, including the **Green Climate Fund (GCF)**, shall serve the Agreement.

Meaning of Adaptation and Mitigation:

Mitigation – Reducing climate change: It involves **reducing the flow of heat-trapping greenhouse gases into the atmosphere**, either by reducing sources of these gases (for example, the burning of fossil fuels for electricity, heat, or transport) or enhancing the “sinks” that accumulate and store these gases (such as the oceans, forests, and soil).

The goal of **mitigation is to avoid significant human interference with Earth's climate, “stabilize greenhouse gas levels in a timeframe sufficient to allow ecosystems to adapt naturally to climate change, ensure that food production is not threatened, and to enable economic development to proceed in a sustainable manner”.**

Adaptation – adapting to life in a changing climate: It involves adjusting to actual or expected future climate. The goal is to reduce our risks from the harmful effects of climate change (like sea-level rise, more intense extreme weather events, or food insecurity). It also includes making the most of any potential beneficial opportunities associated with climate change (for example, longer growing seasons or increased yields in some regions).

- **Carbon Credit**
- A carbon credit is a generic term for any **tradable certificate or permit** representing the right to emit **1 tonne of carbon dioxide or carbon dioxide equivalent**.

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- **Carbon intensity is a measure of how clean our electricity is. It refers to how many grams of carbon dioxide (CO₂) are released to produce a kilowatt hour (kWh) of electricity.**

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How does the Paris Agreement work?

Implementation of the Paris Agreement requires **economic and social transformation**, based on the best available science. The Paris Agreement works on a **five-year cycle** of increasingly ambitious climate action -- or, ratcheting up -- carried out by countries. Since 2020, countries have been submitting their national climate action plans, known as **nationally determined contributions (NDCs)**. Each successive NDC is meant to reflect an increasingly higher degree of ambition compared to the previous version.

Recognizing that accelerated action is required to limit global warming to 1.5°C, the COP27 cover decision requests Parties to revisit and strengthen the **2030 targets** in their NDCs to align with the **Paris Agreement temperature goal** by the end of **2023**, taking into account different national circumstances.

COP 29 HAS CONCLUDED RECENTLY AT BAKU, AJERBAIAJAN, concluded on 24 November with an agreement calling on developed countries to deliver at least **\$300 billion per year** to developing countries **by 2035** to drastically reduce greenhouse gas emissions and protect lives and livelihoods from the worsening impacts of climate change.

- **COP 31 will be held in Antalya, Türkiye, from 9 to 20 November 2026.**

NATIONALLY DETERMINED CONTRIBUTIONS (NDCS):

In their NDCs, countries communicate actions they will take to reduce their greenhouse gas emissions in order to reach the goals of the Paris Agreement.

Long-Term Strategies

To better frame the efforts towards the long-term goal, the Paris Agreement invites countries to formulate and submit **long-term low greenhouse gas emission development strategies (LT-LEDS)**. LT-LEDS provide the long-term horizon to the NDCs. Unlike NDCs, they are **not mandatory**.

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In 2023, the first “global stocktake” will assess progress on Paris Agreement goals and chart a way forward. This process, to be concluded at COP28, will further encourage countries to take ambitious climate actions that keep warming below 1.5 degrees Celsius.

India's target in its Long-Term Low Emission Development Strategy (LT-LEDS) is to achieve **net-zero emissions by 2070**, which involves transitioning to a low-carbon economy across sectors like electricity, transport, and industry, while also enhancing forest cover and promoting sustainable urbanization; this goal is not mandatory like Nationally Determined Contributions (NDCs) but provides a long-term roadmap for climate action.

INDCS (INTENDED NATIONALLY DETERMINED CONTRIBUTIONS):

The INDCs (**Intended Nationally Determined Contributions**) are **plans by governments communicated to the UNFCCC regarding the steps they will take to address climate change domestically**.

As per the COP 19 decision (Warsaw 2013), all Parties were requested to prepare their INDCs, without prejudice to the legal nature of the contributions towards achieving the objectives of the Convention and communicate well in advance of COP 21.

India's INDC

India submitted its INDC to the UNFCCC by early October 2015. It is comprehensive and covers all elements, i.e. **adaptation, mitigation, finance, technology and capacity building**.

India's goal is to reduce the overall **emission intensity** and improve the **energy efficiency** of its economy over time.

It also covers concerns to **protect the vulnerable sectors and segments** of its society.

Therefore, in accordance with the aforesaid provision of the Paris Agreement read with relevant decisions, India hereby communicates an update to its first NDC submitted earlier on October 2, 2015, for the period up to 2030, as under:

NDCs as updated in 2022:

1. To put forward and further propagate a healthy and sustainable way of living based on traditions and values of conservation and moderation, including through a mass

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movement for **'LIFE'– 'Lifestyle for Environment'** as a key to combating climate change

2. **To adopt a climate friendly and a cleaner path** than the one followed hitherto by others at corresponding level of economic development.
3. **To reduce Emissions Intensity of its GDP by 45 percent by 2030, from 2005 level**
4. **To achieve about 50 percent cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030**, with the help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF)
5. **To create an additional carbon sink of 2.5 to 3 billion tonnes of CO₂ equivalent through additional forest and tree cover by 2030.**
6. **To better adapt to climate change** by enhancing investments in development programmes in sectors vulnerable to climate change, particularly agriculture, water resources, Himalayan region, coastal regions, health and disaster management.
7. **To mobilize domestic and new & additional funds from developed countries** to implement the above mitigation and adaptation actions in view of the resource required and the resource gap.
8. **To build capacities, create domestic framework and international architecture for quick diffusion of cutting-edge climate technology** in India and for joint collaborative R&D for such future technologies.

Green Climate Fund (GCF):

- The **Green Climate Fund (GCF)** is a new global fund created to **support the efforts** of developing countries to respond to the challenge of climate change. GCF helps developing **countries limit or reduce their greenhouse gas (GHG) emissions** and adapt to climate change.
- The **decision to set up the Green Climate fund (GCF)** was **taken at COP 16** in Cancun on December 2010 and the GCF was operationalized in COP 17 in Durban in 2011.
- **The GCF is head quartered in Songdo, Incheon City, Republic of Korea.**

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- When the Paris Agreement was reached in 2015, the **Green Climate Fund was given an important role in serving** the agreement and supporting the goal of keeping climate change well below 2 degrees Celsius.
- As per the international agreement, advanced economies should provide an annual assistance of \$100 billion, through public and private sources, by 2020 — the deadline is now extended to 2025.
- In India, NABARD is accredited as National Implementing Entity. NABARD is eligible to submit large size projects having outlay of more than USD 250 million.
- GCF is mandated to invest **50% of its resources** to mitigation and **50% to adaptation in grant equivalent**. At least half of its adaptation resources must be invested in the most climate vulnerable countries (small island developing states (SIDS), least developed countries (LDCs), and African States).

GCF aims to achieve goal by four transitions:

- Built environment; energy & industry;
- Human security,
- Livelihoods and wellbeing;
- Land-use, forests and ecosystems **and**,

Employing a four-pronged approach:

1. Transformational planning and programming
2. Catalyzing climate innovation
3. De-risking investment to mobilize finance at scale
4. Mainstreaming climate risks and opportunities into investment decision

Governing of Fund:

- The **GCF Board** is charged with **the governance and oversight of the Fund's management**. It was established by 194 sovereign governments party to the UN Framework Convention on Climate Change (UNFCCC).
- The **Board is independent** and guided by the Conference of the Parties (COP) to the Convention.
- It is governed by a Board of 24 members and initially supported by a Secretariat.

The World Bank serves as the interim trustee of the GCF, and the Fund functions under the guidance of and remains accountable to the UNFCCC Conference of Parties.

STEPS TAKEN BY INDIAN GOVERNMENT:

1. **Ratifying Paris Agreement:** India has ratified the Paris Agreement with the United Nations on 2 October 2016.

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2. NAPCC: National Action Plan on Climate Change (NAPCC):

The National Action Plan on Climate Change (NAPCC) was released by the Prime Minister on **30th June 2008**.

Government of India is implementing the **National Action Plan on Climate Change (NAPCC)** with a focus **on promoting understanding of climate change** and establishing linkage between adaptation and mitigation consistent with the national priority for achieving sustainable development and to fulfill India's Intended Nationally Determined Contribution (INDC) to the UNFCCC under Paris Agreement.

The Principles of NAPCC are:

- Protecting the poor through an inclusive and sustainable development strategy, sensitive to climate change
- Achieving national growth and poverty alleviation objectives while ensuring ecological sustainability
- Efficient and cost-effective strategies for end-use demand-side management
- Extensive and accelerated deployment of appropriate technologies for adaptation and mitigation
- New and innovative market, regulatory, and voluntary mechanisms for sustainable development
- Effective implementation through unique linkages – with civil society, LGUs, and public-private partnerships

It comprises of eight national missions representing:

1. **National Solar Mission** (under MNRE)
2. **National Mission for Enhanced Energy Efficiency** (under Ministry of Power)
3. **National Mission on Sustainable Habitat** (under Ministry of Housing and Urban Affairs)
4. **National Water Mission** (under MoJS)
5. **National Mission Sustaining the Himalayan Ecosystem** (under MoS&T)
6. **National Mission for a Green India** (under MoEFCC)
7. **National Mission for Sustainable Agriculture** (under Ministry of Agriculture)
8. **National Mission on Strategic Knowledge for Climate Change** (under MoS&T)

An **Executive Committee on Climate Change (ECCC)** under the Chairmanship of Principal Secretary to Prime Minister has been set up for assisting the Prime Minister's Council on Climate Change in evolving a coordinating response to issues relating to climate change with regular monitoring of the eight National missions along with other initiatives on Climate Change and coordinating with various agencies.

1. **National Solar Mission:** The NSM was launched in **January 2010**. The **objective** of the National Solar Mission is to **establish India as a global leader in solar energy**, by creating the policy conditions for its diffusion across the country as quickly as possible. The Mission adopts a **three-phase approach**, Phase 1 (up to 2012 - 13), Phase 2 (2013 - 17) and Phase 3 (2017 - 22). The immediate aim of the Mission is to focus on setting up an

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enabling environment for solar technology penetration in the country both at a centralized and decentralized level.

2. National Mission for Enhanced Energy Efficiency (NMEEE): NMEEE aims to strengthen the **market for energy efficiency** by creating conducive regulatory and policy regime and has envisaged fostering innovative and sustainable business models to the energy efficiency sector. The Mission is implemented since 2011

NMEEE consists of **four initiatives** to enhance energy efficiency in energy intensive industries:

- **Perform, Achieve and Trade (PAT)**
- **Market Transformation for Energy Efficiency (MTEE)**
- **Energy Efficiency Financing Platform (EEFP)**
- **Framework for Energy Efficient Economic Development (FEEED)**

3. National Mission on Sustainable Habitat: The National Mission on Sustainable Habitat was approved by the Prime Minister's Council for Climate Change in June 2010. The key deliverables of the Mission include:

- Development of **sustainable habitat standards** that lead to robust development strategies while simultaneously addressing climate change related concerns
- Preparation of **city development plans** that comprehensively address adaptation and mitigation concerns
- Preparation of **comprehensive mobility plans** that enable cities to undertake long-term, energy efficient and cost-effective transport planning
- **Capacity building** for undertaking activities relevant to the Mission

4. National Water Mission will **ensure integrated water resource management** helping to **conserve water, minimize wastage and ensure more equitable distribution** both across and within states. The Mission will take into account the provisions of the National Water Policy and develop a framework to optimize water use by increasing water use efficiency by 20 per cent through regulatory mechanisms with differential.

5. National Mission for sustaining the Himalayan Eco-system: This particular mission sets the goal to **prevent melting of the Himalayan glaciers** and to protect biodiversity in the Himalayan region.

The mission attempts to address some important issues concerning

- Himalayan Glaciers and the associated hydrological consequences
- Biodiversity conservation and protection
- Wildlife conservation and protection
- Traditional knowledge societies and their livelihood and
- Planning for sustaining the Himalayan Ecosystem

6. National Mission for a Green India (GIM): The Cabinet Committee on Economic Affairs approved a proposal of the Ministry of Environment and Forests for a National Mission for a Green India (GIM) as a Centrally Sponsored Scheme. GIM puts —greening in the context of climate change adaptation and mitigation. **Greening is meant to enhance ecosystem services such as carbon sequestration and storage** (in forests and other ecosystems),

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hydrological services and biodiversity; as well as other provisioning services such as fuel, fodder, small timber and non- timber forest products (NTFPs)

7. National Mission for Sustainable Agriculture (NMSA) has been made operational from the year 2014-15, it aims at making **agriculture more productive, sustainable, remunerative** and climate resilient by promoting location specific integrated /composite farming systems; soil and moisture conservation measures; comprehensive soil health management; efficient water management practices and mainstreaming rain-fed technologies.

8. National Mission on Strategic Knowledge for Climate Change (NMSKCC) seeks to build a vibrant and dynamic knowledge system that would inform and support national action for responding effectively to the objective of ecologically sustainable development

National Adaptation Fund on Climate Change (NAFCC):

- **National Adaptation Fund on Climate Change (NAFCC) was launched in 2015** with an initial outlay of Rs. 350 crores to meet the cost of adaptation to climate change for the State and Union Territories of India that are particularly vulnerable to the adverse effects of climate change.
- The overall aim of the fund is to support concrete adaptation activities which are not covered under on-going activities through the schemes of State and National Government that reduce the adverse effects of climate change facing community, sector and states.
- The Scheme has been taken as Central Sector Scheme with National Bank for Agriculture and Rural Development (NABARD) as the National Implementing Entity (NIE). Besides, enhancing adaptive capacity at national and state level, national conference / workshop, awareness/ information dissemination, Research and Development and establishing a coordination and monitoring unit have also been proposed.

9. State Action Plans on Climate Change (SAPCC) aim to create institutional capacities and implement sectoral activities to address climate change. These plans are focused on adaptation with mitigation as co-benefit in sectors such as water, agriculture, tourism, forestry, transport, habitat and energy. SAPCCs have been endorsed by the National Steering Committee on Climate Change (NSCCC) at the MoEF&CC.

MoEF&CC has also provided financial support to states for enhancing their capacities for undertaking climate change activities.

National Clean Air Programme (NCAP): National Clean Air Programme (NCAP) is a national-level strategy for reducing the levels of air pollution at both the regional and urban scales. Target is for reduction of 20-30% of PM2.5 and PM10 concentration by 2024.

Improve air quality in 131 non-attainment cities.

It is implemented under the guidance of Central pollution control board.

Sustainable Development

In 2022, the government has increased the target to 40% reduction in Particulate Matter by 2026.

NATIONAL AIR QUALITY MONITORING PROGRAMME: Central Pollution Control Board is executing a nation-wide programme of ambient air quality monitoring known as National Air Quality Monitoring Programme (NAMP). The network consists of **manual operating stations covering all the states and UTs in the country.**

Under NAMP, **four air pollutants viz. SO₂, NO₂, suspended particulate matter (PM₁₀), and fine particulate matter (PM_{2.5})** have been identified for regular monitoring at all the locations.

In addition, there are real-time Continuous Ambient Air Quality Monitoring stations (CAAQMS) in various cities across states, monitoring **08 pollutants viz. PM₁₀, PM_{2.5}, SO₂, NO_x, ammonia (NH₃), CO, ozone (O₃), and benzene.**

PM₁₀ are inhalable coarse particles, which are particles with a diameter between 2.5 and 10 micrometers (µm) and **PM_{2.5} are fine particles with a diameter of 2.5 µm or less.**

Particulates are the deadliest form of air pollutants due to their ability to **penetrate deep into the lungs and blood streams unfiltered.** The **smaller PM_{2.5} are particularly deadly as it can penetrate deep into the lungs.**

The monitoring of pollutants is carried out for 24 hours (a 4-hourly sampling for gaseous pollutants and an 8-hourly sampling for particulate matter) twice a week, to have 104 observations in a year.

The monitoring is being carried out with the help of the Central Pollution Control Board (CPCB), State Pollution Control Boards (SPCB), Pollution Control Committees (PCC), National Environmental Engineering Research Institute (NEERI).

NATIONAL AMBIENT AIR QUALITY STANDARDS (NAAQS) Ambient air quality refers to the condition or **quality of the outdoor air.** NAAQs are the standards for ambient air quality with reference to various identified pollutant notified by the CPCB under the Air (Prevention and Control of Pollution) Act, 1981.

INTERNATIONAL SOLAR ALLIANCE (ISA):

Vision: Let us together make the sun brighter.

Mission: Every home no matter how far away, will have a light at home

The **International Solar Alliance (ISA)** is an action-oriented, member-driven, collaborative platform for increased deployment of solar energy technologies **as a means for bringing energy access, ensuring energy security, and driving energy transition in its member countries.**

The ISA strives to develop and deploy cost-effective and transformational energy solutions powered by the sun to help member countries develop low-carbon growth trajectories, with particular focus on delivering impact in countries categorized as Least Developed Countries (LDCs) and the Small Island Developing States (SIDS).

Sustainable Development

Being a global platform, ISA's partnerships with multilateral development banks (MDBs), development financial institutions (DFIs), private and public sector organisations, civil society and other international institutions is key to delivering the change it seeks to see in the world going ahead.

The ISA is guided by its **'Towards 1000' strategy which aims to mobilise USD 1,000 billion of investments in solar energy solutions by 2030**, while delivering energy access to 1,000 million people using clean energy solutions and resulting in installation of 1,000 GW of solar energy capacity. This would help mitigate global solar emissions to the tune of 1,000 million tonnes of CO₂ every year. For meeting these goals, the ISA takes a programmatic approach.

Currently, the ISA has **9 comprehensive programmes**, each focusing on a distinct application that could help scale deployment of solar energy solutions. Activities under the programmes focus on **3 priority areas – Analytics & Advocacy, Capacity Building, and Programmatic Support**, that help create a favourable environment for solar energy investments to take root in the country.

The ISA was conceived as a joint effort by India and France to mobilize efforts against climate change through deployment of solar energy solutions. It was conceptualized on the sidelines of **the 21st Conference of Parties (COP21)** to the United Nations Framework Convention on Climate Change (UNFCCC) held in Paris in 2015. With the amendment of its Framework Agreement in 2020, all member states of the United Nations are now eligible to join the ISA.

The Republic of Armenia has officially become the 104th full member of the International Solar Alliance.

The International Solar Alliance (ISA) is headquartered at the National Institute of Solar Energy (NISE) in Gwalpahari, Gurugram, India.

The Indian government has dedicated five acres of land on the NISE campus for ISA's future headquarters; it also has contributed ₹1.75 billion (US\$27 million) to the fund to build the ISA campus and for meeting expenditures for the ISA's first five years.

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Green bond: A green bond is like any other regular bond but with one key difference: the money raised by the issuer are earmarked towards financing 'green' projects, i.e. assets or business activities that are environment-friendly. Such projects could be in the areas of renewable energy, clean transportation and sustainable water management.

In 2007, green bonds were launched by few development banks such as the European Investment Bank and the World Bank. Subsequently, in 2013, corporates too started participating, which led to its overall growth. Back home, **Yes Bank was the first bank to come out with a issue worth Rs 1,000 crore in 2015.** Following this, few other banks too had green bond issuances. **CLP India, was the first Indian company to tap this route.**

SEBI's indicative list includes renewable and sustainable energy such as wind and solar, clean transportation, sustainable water management, climate change adaptation, energy efficiency, sustainable waste management and land use and biodiversity conservation.

ENVIRONMENT RELATED INITIATIVES BY GOI:

The Ministry of Environment, Forest and Climate Change (MoEFCC) is the nodal agency in the administrative structure of the Central Government for the planning, promotion, co-ordination and overseeing the implementation of India's environmental and forestry policies and programmes.

The Ministry also serves as the nodal agency in the country for the **United Nations Environment Programme (UNEP)**, South Asia Co-operative Environment Programme (SACEP), International Centre for Integrated Mountain Development (ICIMOD) and for the follow-up of the United Nations Conference on Environment and Development (UNCED). The Ministry is also entrusted with issues relating to multilateral bodies such as the Commission on Sustainable Development (CSD), Global Environment Facility (GEF) and of regional bodies like Economic and Social Council for Asia and Pacific (ESCAP) and South Asian Association for Regional Co-operation (SAARC) on matters pertaining to the environment.

The broad objectives of the Ministry are:

- Conservation and survey of flora, fauna, forests and wildlife
- Prevention and control of pollution
- Afforestation and regeneration of degraded areas
- Protection of the environment and
- Ensuring the welfare of animals

These objectives are well supported by a set of legislative and regulatory measures, aimed at the preservation, conservation and protection of the environment. Besides the legislative measures, **the National Conservation Strategy and Policy Statement on Environment and Development, 1992; National Forest Policy, 1988; Policy Statement on Abatement of Pollution, 1992; and the National Environment Policy, 2006** also guide the Ministry's work.

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Compensatory Afforestation Fund Management and Planning Authority (CAMPA): Establishment in 2004, the Ministry of Environment and Forests constituted the **Compensatory Afforestation Fund Management and Planning Authority (CAMPA)** to oversee and manage the Compensatory Afforestation Fund (CAF) as directed by the Supreme Court.

The Hon'ble Supreme Court on 10th July 2009 issued orders that there will be a **Compensatory Afforestation Fund Management and Planning Authority (CAMPA)** as National Advisory Council under the chairmanship of the Union Minister of Environment & Forests & Climate change for monitoring, technical assistance and evaluation of compensatory afforestation activities.

National CAMPA Advisory Council has been established as per orders of The Hon'ble Supreme Court with the following mandate:

- Lay down broad guidelines for State CAMPA.
- Facilitate scientific, technological and other assistance that may be required by State CAMPA.
- Make recommendations to State CAMPA based on a review of their plans and programmes.
- Provide a mechanism to State CAMPA to resolve issues of an inter-state or Centre-State character.

The CAMPA funds are utilised for compensating the **loss of forest land and ecosystem services** by raising of **compensatory afforestation**, improving quality of forests through assisted natural regeneration, enrichment of biodiversity, improvement of wildlife habitat, control of forest fire, forest protection and soil and water conservation measures.

The Compensatory Afforestation Fund (CAF Act), 2016 and Rules, 2018 provide elaborate guidelines and activities for utilization of CAMPA Fund.

NAGAR VAN YOJANA (NVY) scheme by Ministry of Environment, Forest & Climate Change

The total estimated cost of Nagar Van Yojana is Rs.895 crore for the period of 2020-21 to 2024-25 for implementation from the National Funds under Compensatory Afforestation Fund Management and Planning Authority (CAMPA).

Objectives:

1. **Creating green space and** aesthetic environment in an urban set up.
2. Creating **awareness about plants and biodiversity and developing environment stewardship.**
3. Facilitating **in-situ conservation** of important flora of the region.
4. Contributing to environmental improvement of cities by pollution mitigation, providing cleaner air, noise reduction, water harvesting and reduction of heat islands effect.
5. Extending health benefits to residents of the city and
6. Helping cities become climate resilient

General guidelines:

1. **The scheme is proposed to be implemented for a period of five years starting from 2020-21 to 2024-2025.**

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2. The scheme aims **at developing 400 Nagar Vans and 200 Nagar Vatikas** across the country in cities having Municipal Corporation/Municipal Council/Municipalities.

3. **Nagar Van/Nagar Vatika may be developed on forest or other vacant non-forest public land.**

4. The area selected shall be accessible to city dwellers/general public.

5 Educational institutions, Universities, Government/Non-Government organisations/ULB may also take up 'Nagar Van/Vatika' on land owned by them and the proposals are to be routed through State Government.

6. Participation of local people, students in plantations may be encouraged to create a sense of ownership. Innovative concepts like creation of Panchvati, Aushadhi Vatika, Nakshatra Van, Oxyzones etc. may be taken up to attract people in plantation activities.

THE NATIONAL GREEN TRIBUNAL : has been established on **18.10.2010 under the National Green Tribunal Act 2010** (is an Act of the Parliament of India) for effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources including enforcement of any legal right relating to environment and giving relief and compensation for damages to persons and property and for matters connected therewith or incidental thereto.

The Tribunal shall not be bound by the procedure laid down under the Code of Civil Procedure, 1908, but shall be guided by principles of natural justice.

The Tribunal is mandated to make and endeavour for disposal of applications or appeals finally **within 6 months** of filing of the same.

Initially, the NGT is proposed to be set up at five places of sittings and will follow circuit procedure for making itself more accessible. **New Delhi is the Principal Place** of Sitting of the Tribunal and **Bhopal, Pune, Kolkata and Chennai shall be the other 4 place** of sitting of the Tribunal.

The NGT has the power to hear all civil cases relating to environmental issues that are linked to the following laws -

- **The Water (Prevention and Control of Pollution) Act, 1974;**
- **The Forest (Conservation) Act, 1980;**
- **The Air (Prevention and Control of Pollution) Act, 1981;**
- **The Environment (Protection) Act, 1986;**
- **The Public Liability Insurance Act, 1991;**
- **The Biological Diversity Act, 2002.**

There is a bar on civil court to take cases under these listed laws in Schedule 1 of NGT act. The NGT has not been vested with powers to hear any matter relating to the Wildlife (Protection) Act, 1972, the Indian Forest Act, 1927 and various laws enacted by States relating to forests, tree preservation etc.

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ENVIRONMENT AUDIT RULES

Environment audit rules are regulatory provisions that require industries and other entities to systematically evaluate and report their environmental performance.

Main Features of Environment Audit Rules, 2025

- i. Auditors to be **Certified and Registered by MoEFCC notified Environment Audit Designate Agency (EADA)**.
- ii. **EADA to be responsible for** certification and registration of EAs, monitoring their performance, take disciplinary action, facilitating capacity building, maintaining online register.
- iii. **Certification** of Environmental Auditors to be based on **either scrutiny of their qualification and experience or by conducting an exam**.
- iv. **Auditing to be undertaken only by Registered Environment Auditors**.
- v. The assignment of REAs to specific project entities **shall be done by a random assignment method**.
- vi. REAs **responsible for compliance evaluation and related activities of sampling, analysis, compensation calculation, verification** under Green Credit Rules, **audit under waste management rules**, and under various other environment and forest related legislations.
- vii. REAs may also undertake auditing assignment by the PP **including for verification of self compliance report**.

Key Regulatory Stakeholders under the Rules:

- i. **Certified Environment Auditor:** Individuals who qualify through either Recognition of Prior Learning (RPL) or the National Certification Examination (NCE).
- ii. **Registered Environment Auditor:** Those certified individuals officially registered to perform audits.
- iii. **Environment Audit Designated Agency (EADA):** The body responsible for certification, registration, oversight, and training of auditors.
- iv. **Ministry of Environment, Forest and Climate Change;** Oversee the implementation of the Rules and issue necessary guidelines from time to time.
- v. **CPCB/SPCB/ROs:** To continue with their existing role of inspection and verification, as and when required and to assist MoEFCC in overseeing the implementation of the Rules.

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GREEN MOBILITY:

The National Policy on Biofuels 2018, provides an **indicative target of 20% ethanol blending under the Ethanol Blended Petrol (EBP) Programme by 2025.**

Under the new National Policy on Biofuels, the central government has expanded the scope of raw material for ethanol production by allowing use of various agro-waste products. Also advanced vehicle emission and fuel quality standards– BSIV from 2017 and BS-VI from 2020.

Understanding National Air Quality index (AQI):

- AQI is a number used to communicate to the public how polluted the air currently is or how polluted it is forecasted to become.
- As AQI increases, an increasingly large percentage of the population is likely to experience increasingly adverse health effects.
- The AQI is most commonly used by Central Pollution Control Board (CPCB) to describe ground-level ozone levels.
- The classifications of air quality are part of a 6 grade, colour coded taking into account 8 pollutant levels.

AQ sub-index and health breakpoints are evolved for **eight pollutants (PM10, PM2.5, NO2, SO2, CO, O3, NH3, and Pb)** for which short-term (upto 24-hours)

Green Good 0 to 50

Yellow Moderate 51 to 100

Orange Unhealthy for sensitive groups 101 to 150

Red Unhealthy 151 to 200

Purple Very Unhealthy 201 to 300

Maroon Hazardous 301 to 500

UNITED NATIONS DEVELOPMENT PROGRAMME (UNDP): is the United Nations' global development network. **Headquartered in New York City.**

UNDP advocates for change and connects countries to knowledge, experience and resources to help people build a better life.

It provides expert advice, training and grants support to developing countries, with increasing emphasis on assistance to the least developed countries. It promotes technical and investment cooperation among nations.

The status of UNDP is that of an executive board within the United Nations General Assembly. The UNDP Administrator is the third highest-ranking official of the United Nations after the United Nations Secretary-General and Deputy Secretary-General.

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UNDP is based on the merging of the United Nations Expanded Programme of Technical Assistance, created in 1949, and the United Nations Special Fund, established in 1958. **UNDP, as we know it now, was established in 1966 by the General Assembly of the United Nations.**

UNDP's role: The SDGs came into effect in January 2016, and they will continue guide UNDP policy and funding for the next 15 years.

As the lead UN development agency, UNDP is uniquely placed to help implement the Goals through our work in some **170 countries and territories**. Our strategic plan focuses on key areas **including poverty alleviation, democratic governance and peacebuilding, climate change and disaster risk, and economic inequality**. UNDP provides support to governments to integrate the SDGs into their national development plans and policies. This work is already underway, as we support many countries in accelerating progress already achieved under the Millennium Development Goals.

Mission and vision: UNDP's mandate is to end poverty, build democratic governance, rule of law, and inclusive institutions. We advocate for change, and connect countries to knowledge, experience and resources to help people build a better life.

UNDP focuses on helping countries build and share solutions in three main areas:

- **Sustainable development**
- **Democratic governance and peace building**
- **Climate and disaster resilience**

In all our activities, UNDP encourages the protection of human rights and the empowerment of women, minorities and the poorest and most vulnerable. The annual **Human Development Report, commissioned by UNDP**, focuses the global debate on key development issues, providing new measurement tools, innovative analysis and often controversial policy proposals. The global Report's analytical framework and inclusive approach carry over into regional, national and local Human Development Reports, also supported by UNDP.

ENVIRONMENTAL PERFORMANCE INDEX (EPI) :

The **Environment Performance Index (EPI)** is an international ranking system that measures the environmental health and sustainability of countries.

The **EPI, a biennial index**, was started in 2002 as **Environmental Sustainability Index by the World Economic Forum**.

The Environmental Performance Index (EPI) is published by the Yale Center for Environmental Law and Policy and the Center for International Earth Science Information Network, Columbia University.

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Environmental Performance Index

The **2024 EPI** combines **58 indicators across 11 issue categories**, ranging from climate change mitigation and air pollution to waste management, sustainability of fisheries and agriculture, deforestation, and biodiversity protection.

India's rank: 176

India	176	27.6	4.7
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1. Climate change mitigation
2. Air Quality (household solid fuels and PM2.5 exposure)
3. Waste Management
4. Water & Sanitation
5. Heavy Metals (lead exposure)
6. Biodiversity & Habitat
7. Ecosystem services
8. Fisheries
9. Agriculture (sustainable practice)
10. Acid Rain
11. Water resources (wastewater treatment)

WORLD ENVIRONMENT DAY: 5TH JUNE,

The United Nation, aware that the protection and improvement of the human environment is a major issue, which affects the well-being of people and economic development throughout the world, designated 5 June as World Environment Day. The celebration of this day provides us with an opportunity to broaden the basis for an enlightened opinion and responsible conduct by individuals, enterprises and communities in preserving and enhancing the environment. World Environment Day is the biggest annual event for positive environmental action and takes place every 5 June.

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OZONE CELL:

Ozone Layer Protection: Ozone, a tri-atomic molecule of oxygen is formed from oxygen naturally in the upper levels of the Earth's atmosphere by high-energy Ultraviolet (UV) radiation from the Sun. The UV radiation breaks down oxygen molecules, releasing free atoms, some of which bond with other oxygen molecule to form ozone. **About 90 per cent of ozone formed in this way lies between 10 and 50 kilometers above the Earth's surface, called the Stratosphere.** The ozone found in this part of the atmosphere is called the **ozone layer**.

The ozone layer absorbs all the harmful UV-B radiations emanating from the Sun. It protects plant and animal life from UV-B radiation.

The **UV-B radiation has the potential to cause skin cancer, eye cataract, suppress body's immune system, decrease crop yield etc.**, which led to the adoption of the **Vienna Convention for the Protection of the Ozone Layer in 1985** and the **Montreal Protocol on Substances that Deplete the Ozone Layer in 1987**.

The mandate of the Montreal Protocol is to **phase out the production and consumption** of the **Ozone Depleting Substances (ODSs)**.

India is a Party to the Vienna Convention for the Protection of the Ozone Layer and the Montreal Protocol on Substances that Deplete the Ozone Layer and it's all the amendments/adjustments.

Currently HCFCs are being phased out as per the accelerated phase out schedule of the Montreal Protocol.

The MoEF&CC has set up the **Ozone Cell as a National Ozone Unit (NOU)** to render necessary services for effective and timely implementation of the Protocol and its ODS phase-out program in India.

Kigali Amendment to the Montreal Protocol for phase-down of HFCs: The **28th Meeting of Parties to the Protocol held in Kigali in 2016** adopted an amendment to the Protocol which is historic and aimed at phasing down the HFCs that contribute to global warming.

HFCs do not deplete the Ozone layer, however, they have high global warming.

INTERNATIONAL CONVENTIONS/PROTOCOLS:

Stockholm Convention on Persistent Organic Pollutants (POPs) is a **global treaty** to protect human health and the environment **from POPs (Aldrin, Chlordane, DDT, Dieldrin, Endrin etc.)** At present, twenty one chemicals are covered under the Stockholm Convention, of which use of DDT is restricted in India.

Minamata Convention on Mercury: At the Conference of Plenipotentiaries held from **9th-11th October 2013 in Minamata and Kumamoto, Japan**, the **"Minamata Convention on Mercury"**, a global treaty to protect human health and the environment from the adverse effects

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of mercury, was formally adopted and opened for signature by States and regional economic integration organizations. **India has signed the Convention on 30th September 2014.**

Ramsar Convention on Wetland: The Convention on Wetlands, signed in **Ramsar**, Iran, in 1971, is an **intergovernmental treaty which provides the framework for national action and international cooperation** for the conservation and wise use of wetlands and their resources.

The Intergovernmental Panel on Climate Change: Created in 1988 by the World Meteorological Organization (WMO) and the United Nations Environment Programme (UNEP), the **objective of the IPCC is to provide governments at all levels with scientific information that they can use to develop climate policies. IPCC reports are also a key input into international climate change negotiations.**

The IPCC is an organization of governments that are members of the United Nations or WMO. **The IPCC currently has 195 members.** Thousands of people from all over the world contribute to the work of the IPCC. For the assessment reports, IPCC scientists volunteer their time to assess the thousands of scientific papers published each year to provide a comprehensive summary of what is known about the drivers of climate change, its impacts and future risks, and how adaptation and mitigation can reduce those risks. **Headquarter location: Geneva, Switzerland**

The IPCC is divided into three Working Groups and a Task Force.

Working Group I : deals with **The Physical Science Basis of Climate Change**,
Working Group II: with **Climate Change Impacts, Adaptation and Vulnerability**,
Working Group III with **Mitigation of Climate Change**.

The **main objective of the Task Force on National Greenhouse Gas Inventories** is to develop and refine a methodology for the calculation and reporting of national greenhouse gas emissions and removals.

CARBON CREDIT: A carbon credit is a generic term for **any tradable certificate or permit** representing the **right to emit one tonne of carbon dioxide or the equivalent amount** of a different greenhouse gas.

Carbon credits and carbon markets are a component of national and international attempts to mitigate the growth in concentrations of greenhouse gases (GHGs). One carbon credit is equal to one tonne of carbon dioxide, or in some markets, carbon dioxide equivalent gases. Carbon trading is an application of an emissions trading approach. Greenhouse gas emissions are capped and then markets are used to allocate the emissions among the group of regulated sources.

EMISSIONS TRADING: Greenhouse gas emissions a new commodity Parties with commitments under the Kyoto Protocol have accepted targets for limiting or reducing emissions. These targets are expressed as levels of allowed emissions, or assigned amounts. **The allowed emissions are divided into assigned amount units (AAUs).**

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A **Certified Emissions Reduction**, also known as CER, is a certificate issued by the United Nations to member nations for preventing one tonne of carbon dioxide emissions. These are usually issued to member states for projects achieving greenhouse gas reductions through the use of Clean Development Mechanisms (CDM). CDMs make it possible for these projects to occur and set a baseline for future emission targets.

MAJOR GREENHOUSE GASES EMITTING COUNTRIES:

China is the biggest emitter at 26.4% of global greenhouse gas emissions, followed by the United States at 12.5%, India at 7.06%, and the European Union at 7.03%.

Carbon dioxide (CO₂) comprises 74.1% of greenhouse gas emissions.

Most CO₂ emissions (92%) are from the use of fossil fuels, especially for generation of electricity and heat, transportation, and manufacturing and consumption.

Methane (CH₄) make up 17.3% of total greenhouse gas emissions, respectively, mostly from agriculture.

Energy consumption is by far the biggest source of human-caused greenhouse gas emissions.

CLIMATE CHANGE FUND: In keeping with the commitment of NABARD to address impact of climate change the “Climate Change Fund” was created out of the profit of NABARD during 2016-17 for facilitating attempts to address impacts of climate change especially towards fostering sustainable development. NABARD contributes annually from its profit towards the corpus of the fund.

Institution of the “**Climate Change Fund**” is a **unique initiative of NABARD** as a **Development Financial Institution** to foster sustainable development and contribute meaningfully towards national priorities.

- **Objective:** To promote and support activities aimed towards addressing climate change impacts, adaptation and mitigation measures, awareness generation, knowledge sharing and facilitate sustainable development.

UNITED NATIONS ENVIRONMENT PROGRAMME (UN ENVIRONMENT): is the leading global environmental authority that **sets the global environmental agenda, promotes the coherent implementation of the environmental dimension of sustainable development within the United Nations system**, and serves as an authoritative advocate for the global environment. It was founded by Maurice Strong, its first director, as a result of the United Nations Conference on the Human Environment (Stockholm Conference) in June 1972 .

The United Nations Environment Programme (UNEP) is the leading global authority on the environment.

UNEP's mission is to **inspire, inform, and enable nations and peoples** to improve their quality of life without compromising that of future generations.

UNEP is here to lead this transformation with its 193 Member States, and in partnership with other UN agencies and stakeholders.

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The Environment Fund is UNEP's core source of flexible funds, providing the bedrock of the work worldwide.

In addition, earmarked funds (funds given or "earmarked" to a specific project, theme, country etc.) enable UNEP to expand and replicate the programme in more countries and with more partners. **Main providers of earmarked funds include the Global Environment Facility, the Green Climate Fund and the European Commission.**

Headquartered in Nairobi, Kenya

The Environment Fund, established already in 1973 by the UN General Assembly, is the core financial fund of the UN Environment Programme (UNEP).

Since 2012, UNEP membership encompasses all 193 UN Member States, who are responsible for funding the programme.

- **The Environment Fund is used for:**

- Implementation of the Medium-Term Strategy and its programmes;
- Keeping the environment under review. For example, flagship scientific publications such as the Global Environment Outlook, the Global Chemicals Outlook and the Emissions Gap Report translate the best available scientific knowledge into information relevant for decision makers, who then can turn policy into action;
- Identification of new emerging environmental issues

Green Good Deeds, the societal movement launched by the Union Minister for Environment, Forest & Climate Change, **Dr Harsh Vardhan**, to protect environment and promote good living in the country, has found acceptance by the global community.

COP 29 : TOOK PLACE IN BAKU, AZERBAIJAN FROM 11 TO 22 NOVEMBER 2024

It offered an important opportunity for governments and other global stakeholders to put in place the **mechanisms, tools and signals** required for countries to continue implementing their contributions to the Paris Agreement as urgently needed. The main areas of negotiation were on:

- The **design and key elements of the future climate financial architecture** and the **next collective quantified goal (NCQG) on climate finance**;
- The **rulebook for the future international, compliance carbon markets**; and
- Countries' collective signaling on mitigation ambition, as follow up to the various energy transition objectives captured in last year's COP28 'UAE Consensus'.

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COP 30

The 30th Conference of Parties under the UNFCCC (COP30) concluded in Belém, Brazil, where nations adopted the **Belém Package** as the main negotiated outcome.

What are the Key Outcomes of COP30?

Belém Package

A set of **29 agreed** decisions aimed at shifting focus from discussions to execution, with emphasis on finance, just transition, adaptation monitoring, gender inclusion, and stronger collaboration to achieve Paris Agreement goals.

The summit was described as the “COP of Implementation,” “COP of Truth,” and the “Forest COP.”

- **Global Mutirão Agreement**

COP30 concluded with this agreement prioritising cooperation and implementation over new binding targets.

It reflects a middle ground between developed and developing nations, focusing more on feasibility than ambition.

- **Global Mutirão Platform**

Brazil introduced a digital platform to promote collective climate efforts and bridge the gap between promises and action, especially in energy, finance, and trade sectors.

- **Just Transition Mechanism (JTM / BAM)**

A new mechanism to assist workers and economies in moving away from fossil fuels through capacity-building and cooperation, though it lacks assured financial backing.

- **Implementation Tracker & Mission to 1.5°C**

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These tools aim to track whether countries' actions and **NDCs** align with limiting global warming to 1.5°C, highlighting a shift towards monitoring implementation rather than setting fresh targets.

- **NAP Implementation Alliance**

Launched to accelerate national adaptation planning.

Countries agreed to **triple adaptation finance by 2030 (from 2025 levels)**, but uncertainty remains regarding funding sources.

- **Global Goal on Adaptation (GGA)**

Countries finalised the Baku Adaptation Roadmap and adopted **59 voluntary indicators** to measure progress under adaptation goals.

- **Belém Health Action Plan**

Introduced on Health Day (13 Nov 2025), it aims to strengthen global health systems to tackle climate impacts, based on equity, justice, and community participation.

- **Tropical Forests Forever Facility (TFFF)**

A Brazil-led initiative that rewards countries financially for conserving tropical forests using satellite monitoring, targeting mobilisation of about **USD 125 billion**.

- **Belém 4x Pledge**

Aims to increase the use of sustainable fuels **fourfold by 2035** (compared to 2024), monitored by the International Energy Agency, focusing on fuels like hydrogen and biofuels.

- **Declaration on Hunger & Poverty**

Signed by 43 countries and the EU, it prioritises vulnerable populations in climate policy, focusing more on adaptation through social protection and resilience-building measures.

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- **Belém Gender Action Plan (GAP)**

Focused on strengthening gender-sensitive climate policies and ensuring active participation of women, particularly from vulnerable groups, in climate governance.

KEY TAKEAWAYS:

- **New global climate finance goal:** It calls on “all actors” to raise at least \$1.3 trillion per year, and on developed countries to lead the mobilization of at least \$300 billion, by 2035.
- It launched a “**Baku-Belem Roadmap to 1.3T**”—a conversation to scale up finance in 2025.
- **Non-market approaches (NMAs) (Article 6.8)** This decision is about the implementation of a work programme on NMAs (such as actions addressing biodiversity and climate, and/or enhancing value systems).
- The next phase will consider ways to overcome barriers to NMAs, to use NMAs to implement NDCs, and to capture them on the new online platform, based on inputs from countries and observers by March 31, 2025.
- **Indigenous peoples platform** (LCIPP): A new 2025–2027 workplan for this platform was approved, and the secretariat will explore options for enhancing the participation of Indigenous representatives and for providing interpretation for Indigenous languages.
- **Gender and climate change** : The decision noted that gender balance has either stayed the same or declined across the UNFCCC and has been inconsistent, but that country delegations have been gender-balanced recently. **The decision extends the Lima Work Programme on Gender for 10 years and launches** a new gender action plan, inviting proposals for its scope, and encourages Parties to appoint gender focal points, enhance reporting and integrate gender in climate finance.

THE GREEN CREDIT RULES, 2023 BY MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE:

The **Indian Council of Forestry Research and Education** (ICFRE) is the **administrator** of the programme.

Objectives of Green Credit programme:

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- **The green credit programme shall incentivise environmental positive actions** through market-based mechanism and generate green credit, **which shall be tradable and made available for trading on a domestic market platform.**
- The green credit will arise from **taking measures by a person of any environment activities.**
- The green credit programme shall **encourage industries, companies and other entities to meet their existing obligations or other obligations under any law for the time being in force, and encourage other persons and entities,** to undertake voluntary environmental measures by generating or buying green credit:

The Green Credit program encompasses **measures which includes following eight key types of activities** aimed at enhancing environmental sustainability:

- (i) tree plantation**—to promote activities for increasing the green cover across the country;
- (ii) water management**—to promote water conservation, water harvesting and water use efficiency or water savings, including treatment and reuse of wastewater;
- (iii) sustainable agriculture**—to promote natural and regenerative agricultural practices and land restoration to improve productivity, soil health and nutritional value of food produced;
- (iv) waste management**—to promote circularity, sustainable and improved practices for waste management, including collection, segregation, and environmentally sound management;
- (v) air pollution reduction**—to promote measures for reducing air pollution and other pollution abatement activities;
- (vi) mangrove conservation and restoration**—to promote measures for conservation and restoration of mangroves;
- (vii) ecomark label development**—to encourage manufacturers to obtain ecomark label for their goods and services;
- (viii) sustainable building and infrastructure**—to encourage the construction of sustainable buildings and other infrastructure using environment friendly technologies and materials

Important highlights of The Green Credit Rules, 2023:

- **Green Credit” means a singular unit of an incentive provided for a specified activity, delivering a positive impact on the environment.**
- A person or **entity desirous of obtaining green credit** shall register the activity with the Administrator for any activity referred above.
- An **application for registration shall be made to the Administrator electronically** through a website established by the Central Government for that purpose.
- **On receipt of the application, the Administrator (ICFRE)** shall cause the activity to be verified by a designated agency.
- The designated agency shall, **after making such verification and inquiry as it may deem necessary, in accordance with the guideline made in this behalf, submit a report** to the Administrator verifying the activities undertaken by the applicant.
- On receipt of the report verifying the activities undertaken by the applicant, the Administrator shall grant the applicant a certificate of green credit.

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- The methodology for evaluation and verification of the activities undertaken for the purpose of calculation of green credit shall be such as may be determined by the Administrator.
- The Administrator shall seek approval of the Central Government for guidelines and methodologies.
- **A Steering Committee to be constituted by the Central Government shall be responsible** for the monitoring of the implementation of the Green Credit programme under these rules.
 - The **Steering Committee shall comprise of representatives from the Ministries or Departments, experts from the field of environment, industry associations and other relevant stakeholders as the Central Government may consider appropriate**
- The Central Government, based on the recommendations of the Administrator, may constitute **Technical Committees** for each activity which shall assist the Administrator in implementation of the Green Credit programme under these rules.
 - **Each Technical Committee** shall comprise of members from the Ministries and Departments, organisations and experts having the knowledge and experience in the field related to the activity.
- **Green Credit Registry:**
 - The Administrator or designated agency shall establish and maintain a Green Credit Registry **for the registration and issuance of each Green Credit.**
 - The Registry shall be an electronic database, which, inter alia, shall contain common data elements relevant to the registration and issuance of green credit.
 - The Registry shall discharge the following functions, namely:
 - (a) registration of activities and issuance of green credit;
 - (b) ensure accurate accounting of the issuance of green credit;
 - (c) maintain secure database with all required and essential security protocols;
 - (d) any other function assigned to it by the Administrator
- **Trading platform:**
 - (1) The Administrator shall establish and maintain a trading platform with the approval of the Central Government.
 - (2) The trading platform shall perform functions regarding the trading of green credit, in accordance with the guidelines made by the Administrator with the approval of the Central Government.
- **The activities** of the Administrator, designated agency, Registry, trading platform and knowledge and data platform shall be **audited within a period of one year at the end of every third financial year by independent auditors to be appointed by the Central Government on the recommendation of the Steering Committee.**
- The Administrator shall submit an action taken report on **the audit report to the Central Government within a period of six months** from the date of the receipt of the audit report.

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ECOMARK SCHEME BY MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE UNDER ENVIRONMENT (PROTECTION) ACT, 1986:

The Central Pollution Control Board administers the Ecomark Scheme in partnership with Bureau of Indian Standards (BIS), which is the national body for standards and certification.

Ecomark Scheme inter-alia provided **a tool to the consumers to pursue sustainable consumption patterns** as well as to the industry to implement environment-friendly processes or production methods.

Introduction –

- The **Ecomark Certification Rules (hereinafter referred to as ‘the Ecomark Rules’)** is **for labelling of products which will have lesser adverse impacts on the environment**, with the objective to encourage the consumers to adopt such products as well as the manufacturers for transitioning to production of Ecomark certified products for promoting sustainability
- **The Ecomark Rules will promote environmentally friendly products** and ensure environmental performance of such products w.r.t. resource consumption and environmental impacts, in particular the impact on climate change, the impact on nature and biodiversity, energy consumption, generation of waste, emissions to all environmental media, pollution through physical effects and use and release of hazardous substances
- The Ecomark Rules will provide labelling to products that meet approved environmental criteria.

Note: The Bureau of Indian Standards (BIS) and the Central Pollution Control Board (CPCB) jointly implement the Ecomark scheme in India.

Objectives –

(1) The **Ecomark Rules are intended to encourage the demand for environmental friendly products** that cause lesser adverse impacts on the environment thereby supporting the principles of ‘LiFE (Lifestyle for Environment)’, promoting resource efficiency & circular economy and preventing misleading information on environmental aspects of products.

(2) The overall objectives of the Ecomark Rules are as follows:-

- build consumer awareness on environmental issues and of the implications of their choices, thereby generating a change towards more environmental friendly behaviour and consumption patterns;
- encourage manufacturers for transitioning to production of Ecomark certified products;
- prevent misleading and deceptive information with respect to fraudulent use of Ecomark label.

Important definitions:

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- **Ecomark Certificate'** means a certificate issued under the Ecomark Rules by the Administrator authorising a person or a body of persons to mark its product with the Ecomark;
- **'Designated Ecomark Verifier'** means an agency/person designated to carry out verification of the product against set criteria;
- **'Ecomark Certificate Holder'** means a business operator authorized by an Ecomark Administrator to use Ecomark label;
- **'Extended Producer Responsibility (EPR)'** means an environmental strategy in which a producer's responsibility for a product is extended to the post-consumer stage of a product's life cycle especially for recycling, and disposal of their products once those products are designated as no longer useful by consumers.
- **Criteria for Ecomark –**
 - (1) **Environmental criteria for each product /product category under Ecomark Rules shall be notified by the Central Government.** The criteria shall be for broad environmental levels and aspects, with specific criteria at the product level.
 - (2) Certification of Standards Body (national or international) for quality and safety, or mandate of Quality Control Orders shall be pre-requisite for Ecomark.

Product primary criteria shall inter-alia include, but not limited to the following –

- i. Production or process including source of raw material;
- ii. Use of natural resources;
- iii. Likely impact on the environment;
- iv. Effect & extent of emissions/waste arising from the production process;
- v. Disposal of the product and its packaging;
- vi. Compliance to Extended Producer Responsibility regulations (where applicable);
- vii. Utilization of waste and recycled materials;
- viii. Suitability for recycling;
- ix. Substitution of hazardous substances with safer ones.

The Central Government will administer the Ecomark Rules with an objective to enhance the efforts for protecting and conserving the environment in the country. The governance of the Ecomark Rules for its effective implementation **shall vest in the Steering Committee**. The Steering Committee will comprise representatives from the concerned Ministries/Departments, domain experts, representatives from industry associations, consumer groups and other relevant stakeholders.

The Central Pollution Control Board shall be the Administrator of the Ecomark Rules and shall be responsible for implementation of the Ecomark Rules.

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The Administrator may constitute **Technical Committees** comprising of members having requisite knowledge from the specified categories, representatives from concerned Ministries/Departments, subject matter experts, representatives from industry associations and other stakeholders;

Ecomark Portal: This Portal would function under the supervision of the Administrator

The Administrator by itself or through Designated Ecomark verifiers shall verify compliance with ecolabelling criteria for the award/renewal of certificate to products under the Ecomark Rules. Designated Ecomark Verifiers shall be accredited entities.

Market surveillance shall inter-alia include picking samples from market, keeping an eye on misuse of Ecomark, raiding suspected places etc. It shall be different from the activities to be undertaken by Designated Ecomark verifiers.

- The Administrator shall by itself or through its empanelled third-party market surveillance agencies, evaluate that the product for compliance according to the certification, on a regular basis.
- Market surveillance third party agencies shall file annual returns in the prescribed form on the Ecomark Portal on or before 31st May of the succeeding year, to which the return relates.

The State Pollution Control Board (SPCB)/ Pollution Control Committee (PCC) shall create awareness about the Ecomark Rules through media, publications, advertisements, posters or by such other means of communication with the support of Local Bodies;

The Central Government on recommendation of the Steering Committee may incentivize Ecomark through any program related to protection and conservation of environment.

Implementation Committee: A Committee shall be constituted by the Central Government under chairpersonship of Chairman, Central Pollution Control Board to recommend measures to Steering Committee for effective implementation of the Ecomark Rules.

- The Committee shall comprise of representatives of the concerned Ministries/Departments, representatives of industry associations and other relevant stakeholders. The Chairperson of the Committee may co-opt any stakeholder/expert to this Committee;

Carbon Credit Trading Scheme, 2023 by Ministry of Power:

- India's Carbon Credit Trading Scheme, 2023 (CCTS 2023) was notified by the Government of India on 28 June 2023 under the Energy Conservation Act, 2001, **to develop the country's first-ever domestic carbon market.**
- The **market will be driven by setting Greenhouse Gas (GHG) emission intensity reduction targets** in line with India's Nationally Determined Contributions (NDC) for selected entities to be obligated under CCTS 2023.

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- The CCTS 2023 entails the **formation of a National Steering Committee or Indian Carbon Market (NSCICM) for the governance and direct oversight of the ICM.**
- **The committee will be chaired by the Secretary; Ministry of Power (MoP);** and co-chaired by the Secretary, Ministry of Environment, Forests and Climate Change (MoEF&CC).
- **The Bureau of Energy Efficiency (BEE)** will be the administrator for the ICM and will be responsible for the development of the GHG emissions trajectory and the targets for the entities to be obligated under the notification.
- **The Grid Controller of India Limited** will be the designated agency for the maintenance of the ICM Registry and will register the obligated entities and maintain the record of the transactions among the obligated entities, among other functions.
- **The Central Electricity Regulatory Commission (CERC)** will be the regulator for the **trading of carbon credit certificates.** They will safeguard the interests of the buyers and the sellers, decide on the frequency of trading, and take action to prevent fraud or mistrust. **The CERC will register the power exchanges to trade the carbon credit certificates** and decide on and notify the rules of trading periodically.

PARIVESH:

PARIVESH is a web based, role-based workflow application which has been developed for **online submission and monitoring of the proposals submitted** by the proponents for **seeking Environment, Forest, Wildlife and CRZ Clearances** from Central, State and district level authorities.

To enhance user experience on **PARIVESH** with modern day web application, Ministry has expanded the scope of existing **PARIVESH (2.0)** leveraging on emerging technology such as **GIS, Advance Data Analytics, etc.** for providing **faster decisions on Green Clearances** and **robust compliance monitoring by end-to-end online appraisal and grant of clearances.**



Major modules in PARIVESH 2.0 viz: end to end online processing of Category A and B environmental clearance proposals was developed and rolled out at Central and STATE-level, respectively.

In addition, **all major functionalities of other major clearances (FC/WL&CRZ) have been developed and rolled out.** In CRZ clearance, all nine State Coastal Zonal Management Authority were onboarded, for the first time, on PARIVESH 2.0 for online submission and

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processing of applications. Furthermore, **PARIVESH 2.0** is also integrated with **Gatishakti and National Single window portal**.

Note: From July 1, 2022, India banned single-use plastic items that have low utility but are frequently littered around, such as plastic straws.

India's 4th Biennial Update Report (BUR-4) to the United Nations Framework Convention on Climate Change (UNFCCC) was submitted on **30th December, 2024**. According to which, following are key achievements:

India's achievements in respect of the NDC targets:

- India has progressively continued decoupling economic growth from GHG emissions. Between **2005 and 2020, India's emission intensity of Gross Domestic Product (GDP) reduced by 36%**.
- By October 2024, the share of non-fossil sources in the installed electricity generation capacity was 46.52%. **Total installed capacity of renewable power, including large hydropower, is 203.22 GW** and cumulative renewable power installed capacity (excluding large hydro projects) has increased 4.5 times from 35 GW in March 2014 to 156.25 GW.
- **India's forest and tree cover has consistently increased and currently stands at 25.17%** of the total geographical area of the country. During 2005 to 2021, additional carbon sink of 2.29 billion tonnes of CO₂ equivalent has been created.

At **COP 26**, India underlined the need to start the one-word movement 'LIFE' which means '**Lifestyle For Environment**' urging for mindful and deliberate utilization instead of mindless and destructive consumption of natural resources. The Hon'ble Prime Minister of India in Glasgow in November 2021 further enhanced the ambition on addressing climate. **These include five nectar elements (Panchamrit) of India's climate action**

- 1. Reach 500GW non-fossil energy capacity by 2030**
- 2. 50 per cent of its energy requirements from renewable energy by 2030**
- 3. Reduction of total projected carbon emissions by one billion tonnes from now to 2030**
- 4. Reduction of the carbon intensity of the economy by 45 per cent by 2030, over 2005 levels**
- 5. Achieving the target of net zero emissions by 2070**

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In keeping with the ambition to significantly reduce the carbon intensity of the economy, the Union Budget 2022-23 announced the issue of Sovereign Green Bonds:

‘As a part of the government’s overall market borrowings in 2022-23, sovereign Green Bonds will be issued for mobilizing resources for green infrastructure. The proceeds will be deployed in public sector projects which help in reducing the carbon intensity of the economy.’

Sovereign Green Bond Framework:

Department of Economic Affairs, Ministry of Finance reserves the right to modify this Framework according to international best practices or in accordance with the Government of India’s international commitments and environmental priorities.

The framework is designed to comply with **four components and key recommendations of the International Capital Market Association (ICMA) Green Bond Principles (2021)**

The four core components as outlined by ICMA green bond principles are:

- i. Use of proceeds;
- ii. Project evaluation and selection;
- iii. Management of proceeds; and
- iv. Reporting.

A ‘green project’ classification is based on the following principles:

- Encourages energy efficiency in resource utilization
- Reduces carbon emissions and greenhouse gases
- Promotes climate resilience and/or adaptation
- Values and improves natural ecosystems and biodiversity especially in accordance with SDG principles

Government of India will use the proceeds raised from Sovereign Green Bonds (SGrB) to finance and/or refinance expenditure (in parts or whole) for eligible green projects in 9 categories:

1. Renewable Energy
2. Energy Efficiency
3. Clean Transportation
4. Climate Change Adaptation
5. Sustainable Water and Waste Management
6. Pollution Prevention and Control
7. Green Buildings
8. Sustainable Management of Living Natural Resources and Land Use
9. Terrestrial and Aquatic Biodiversity Conservation

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All eligible Green Expenditures will include public expenditure undertaken by the Government in the form of investment, subsidies, grants-in-aid, or tax foregone (or a combination of all or some of these) or select operational expenditures, R&D expenditures in public sector projects that help in reducing the carbon intensity of the economy and enable country to meet its Sustainable Development Goals (SDGs).

Equity is allowed only in the sole case of metro projects under 'Clean Transportation' category, as all metro projects in the country access the public share of their total funding through equity investments. Metro projects are implemented through Special Purpose Vehicles (SPVs) meant exclusively for metro transportation projects, and no other project related funding are permissible.

The eligible expenditures are limited to government expenditures that occurred maximum 12 months prior to issuance.

It will be endeavoured that all the proceeds get allocated to projects within 24 months following issuance.

Any expenditure already financed and/or refinanced by dedicated source/ other Government agency will be excluded to ensure suitable oversight and avoid double-counting.

Excluded Projects:

- Projects involving new or existing extraction, production and distribution of fossil fuels, including improvements and upgrades; or where the core energy source is fossil-fuel based
- Nuclear power generation
- Direct waste incineration
- Alcohol, weapons, tobacco, gaming, or palm oil industries
- Renewable energy projects generating energy from biomass using feedstock originating from protected areas
- Landfill projects
- Hydropower plants larger than 25 MW

Expenditures directly related to fossil fuel are excluded. However, investments/expenditures aimed at a relatively cleaner Compressed Natural Gas (CNG) is allowed as an 'eligible expenditure' when used in public transportation projects only. Subsidy/incentive for private transportation using CNG is neither envisaged nor included

Green Finance Working Committee" (GFWC): Ministry of Finance has constituted a "Green Finance Working Committee" (GFWC) with representation from relevant line ministries and chaired by Chief Economic Adviser, Government of India. GFWC will meet at least twice a year to support Ministry of Finance with selection and evaluation of projects and other relevant work related to the Framework. Initial evaluation of the project will be the responsibility of the concerned Ministry/Department in consultation with experts.

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GFWC has been established with clear lines of authority to oversee and validate key decisions on issuance of Sovereign green bonds under the Chairmanship of Chief Economic Adviser, Government of India and members from implementing departments, Ministry of Environment, Forests and Climate Change (MoEFCC), Niti Aayog (India's premier public policy think-tank), Budget Division of Department of Economic Affairs (DEA) and Infrastructure Finance Secretariat, DEA.

For every successive year, GFWC will meet to identify a fresh set of eligible expenditures in line with the Framework in consultation with Line Ministries.

Ministries that are currently not part of GFWC will also be consulted for generating awareness towards sustainability and green projects within their purview, to identify new projects

Principles of selection of green projects by GFWC:

- **Eligible green expenditures are mapped to the environmental objectives** of ICMA Green Bond principles, UN Sustainable Development Goals (SDGs)
- **Eligible Green Expenditures must have** alignment with the objectives of National Conservation Strategy and Policy Statement on Environment and Development, 1992; National Forest Policy, 1988; Policy Statement on Abatement of Pollution, 1992; and the National Environment Policy, 2006 Additionally all such expenditures will adhere to minimum social safeguards as accorded by the Constitution as well as the laws of the country (such as The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013)

Management of Proceeds: The **proceeds will be deposited to the Consolidated Fund of India** (CFI) in line with the regular treasury policy, and then funds from the CFI will be made available for eligible green projects.

For the purposes of ensuring that the proceeds' allocation and accounting is transparent, clear and beyond doubt, a separate account will be created and maintained by the Ministry of Finance Government of India.

Public Debt Management Cell (PDMC) will **keep a track of proceeds within the existing guidelines regarding debt management, and monitor the allocation of funds towards eligible green expenditures.**

Unallocated proceeds, if any, will be carried forward to successive years for investment in eligible green projects only.

It will be endeavoured that all proceeds are allocated within a span of two years from the date of issuance.

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Ministry of Finance will set up a dedicated information system with a view to maintaining a complete **Green Register** including the details of the green bond issuance, proceeds generated, allocations made to eligible projects including information about the eligible projects (summary of the project details, allocation of proceeds to each project, expected climate impact and the extent of unallocated proceeds, both aggregate as well as project-wise).

An amount at least equal to the net proceeds from the completed issuances under this Framework will be allocated to the financing and/or refinancing of expenditures that meet the eligibility criteria.

Allocation and utilization of Green Bonds is also under the purview of audit by Comptroller and Auditor General (CAG)

External Review: This Green Bond Framework has been reviewed by CICERO which has issued an independent Second Party Opinion. It has approved the alignment of the Sovereign Green Bond Framework of the Government of India with the ICMA Green Bond Principles.

It rated India's green bond framework as 'medium green' with 'good' governance. The framework lists **nine broad categories of eligible** green projects to fulfil various environmental objectives such as reducing greenhouse gas emissions, encouraging energy efficiency, promoting climate adaptations and improving natural ecosystems.

Centre for International Climate Research (CICERO), Norway is an interdisciplinary climate research institute established in 1990. CICERO is a world-leading institute which delivers high-quality research that helps society to respond to the climate challenge and strengthen international climate cooperation

Framework for acceptance of Green Deposits:

- *All Scheduled Commercial Banks including Small Finance Banks (excluding Regional Rural Banks, Local Area Banks and Payments Banks) All Deposit taking Non-Banking Finance Companies (NBFCs) including Housing Finance Companies (HFCs)*
- **To encourage regulated entities (REs)** to offer green deposits to customers, protect interest of the depositors, aid customers to achieve their sustainability agenda, address greenwashing concerns **and help augment the flow of credit to green activities/projects.**
- **“green deposit”** means an interest-bearing deposit, received by the RE for a fixed period and the proceeds of which are earmarked for being allocated towards green finance;

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- **green finance** means lending to and/or investing in the activities/projects meeting the requirements prescribed of these guidelines that contributes to climate risk mitigation, climate adaptation and resilience, and other climate-related or environmental objectives - including biodiversity management and nature-based solutions;
- **greenwashing** means the practice of marketing products/services as green, when in fact they do not meet requirements to be defined as green activities/projects.
- REs shall issue green deposits as cumulative/non-cumulative deposits. On maturity, the green deposits would be renewed or withdrawn at the option of the depositor. **The green deposits shall be denominated in Indian Rupees only.**
- REs shall put in place a comprehensive **Board-approved policy on green deposits** laying down therein, all aspects in detail for the issuance and allocation of green deposits. A copy of the above policy on 'Green Deposits' shall be made available on the website of the RE.
- **The eligible green activities/project:**
 1. Renewable Energy
 2. Energy Efficiency
 3. Clean Transportation
 4. Climate Change Adaptation
 5. Sustainable Water and Waste Management
 6. Pollution Prevention and Control
 7. Green Buildings
 8. Sustainable Management of Living Natural Resources and Land Use
 9. Terrestrial and Aquatic Biodiversity Conservation

Exclusions:

- Projects involving new or existing extraction, production and distribution of fossil fuels, including improvements and upgrades; or where the core energy source is fossil-fuel based
- Nuclear power generation.
- Direct waste incineration.
- Alcohol, weapons, tobacco, gaming, or palm oil industries.
- Renewable energy projects generating energy from biomass using feedstock originating from protected areas.
- Landfill projects.
- Hydropower plants larger than 25 MW

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NATIONAL GREEN HYDROGEN MISSION:

Implemented by Ministry of New and Renewable Energy:

India has set its sight on becoming energy independent by 2047 and achieving Net Zero by 2070. To achieve this target, increasing renewable energy use across all economic spheres is central to India's Energy Transition. **Green Hydrogen is considered a promising alternative for enabling this transition. Hydrogen can be utilized for long-duration storage of renewable energy,** replacement of fossil fuels in industry, clean transportation, and potentially also for decentralized power generation, aviation, and marine transport.

Energy use has doubled in the last 20 years and is likely to grow by at least another 25% by 2030. India currently imports over 40% of its primary energy requirements, worth over USD 90 billion every year

The National Green Hydrogen Mission was approved by the Union Cabinet on 4 January 2023 with an outlay of ₹ 19,744 crore from FY 2023-24 to FY 2029-30, with the intended objectives of:

- Making India a leading producer and supplier of Green Hydrogen in the world
- Creation of export opportunities for Green Hydrogen and its derivatives
- Reduction in dependence on imported fossil fuels and feedstock
- Development of indigenous manufacturing capabilities
- Attracting investment and business opportunities for the industry
- Creating opportunities for employment and economic development
- Supporting R&D projects

The initial outlay for the Mission will be 19,744 crore, including an outlay of 17,490 crore for the SIGHT programme, 1,466 crore for pilot projects, 400 crore for R&D, and 388 crore towards other Mission components

The following components have been announced as part of the Mission:

- Facilitating demand creation through exports and domestic utilization;
- Strategic Interventions for Green Hydrogen Transition (SIGHT) programme, which includes incentives for manufacturing of electrolyzers and production of green hydrogen;
- Pilot Projects for steel, mobility, shipping etc.;
- Development of Green Hydrogen Hubs;
- Support for infrastructure development;
- Establishing a robust framework of regulations and standards;
- Research & Development programme;
- Skill development programme; and
- Public awareness and outreach programme.

MISSION OUTCOMES

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The mission outcomes **projected by 2030** are:

- Development of **green hydrogen production capacity of at least 5 MMT** (Million Metric Tonne) per annum with an associated renewable energy capacity addition of about 125 GW in the country
- **Over Rs. Eight lakh crore in total investments**
- **Creation of over Six lakh jobs**
- **Cumulative reduction in fossil fuel imports over Rs. One lakh crore**
- Abatement of nearly 50 MMT of annual greenhouse gas emissions

TYPES OF HYDROGEN BASED ON EXTRACTION METHODS

Depending on the nature of the method of its extraction, hydrogen is categorised into three **categories, namely, Grey, Blue and Green.**

- **Grey Hydrogen:** It is **produced via coal or lignite gasification** (black or brown), or via a process called steam methane reformation (SMR) of natural gas or methane (grey). These tend to be **mostly carbon-intensive processes.**
- **Blue Hydrogen:** It is produced via **natural gas or coal gasification combined with carbon capture storage (CCS) or carbon capture use (CCU)** technologies to reduce carbon emissions.
- **Green Hydrogen:** It is **produced using electrolysis of water with electricity generated by renewable energy.** The carbon intensity ultimately depends on the carbon neutrality of the source of electricity (i.e., the more renewable energy there is in the electricity fuel mix, the "greener" the hydrogen produced).

Ministry of New and Renewable Energy (MNRE) will be responsible for overall **coordination and implementation of the Mission.** The Mission Secretariat, headquartered in MNRE, will formulate schemes and programmes for financial incentives to support production, utilization and export of Green Hydrogen and its derivatives

Ministry of Power (MoP) will implement policies and regulations to ensure delivery of renewable energy for Green Hydrogen production at least possible costs including through development of the necessary power system infrastructure.

Ministry of Petroleum and Natural Gas (MoPNG) will facilitate uptake of Green Hydrogen in refineries and city gas distribution through both Public Sector Entities and private sector.

MISSION GOVERNANCE FRAMEWORK: An Empowered Group (EG) chaired by the **Cabinet Secretary** and comprising Principal Scientific Adviser to the Government of India, CEO, NITI Aayog, and Secretaries of Ministries of New and Renewable Energy, Petroleum and Natural Gas, Power, Road Transport and Highways, Steel, Heavy Industries, Ports, Shipping and Waterways, Skill Development and Entrepreneurship;

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and Departments of Fertilizers, Science and Technology, Scientific and Industrial Research, Promotion of Industry and Internal Trade; and experts from the industry will be set up. The EG will oversee the Mission activities, provide guidance, continuously monitor progress, recommend policy interventions to be made in furtherance of mission objectives

Mission LiFE is an India-led global mass movement to nudge individual and community action to protect and preserve the environment. At the **26th session of the Conference of the Parties (COP26) to the United Nations Framework Convention on Climate Change (UNFCCC) held in Glasgow, United Kingdom**, India shared the mantra of LiFE - **Lifestyle for Environment** - to combat climate change. India is the first country to include LiFE in its Nationally Determined Contributions (NDCs).

It is estimated that global economy could lose up to 18% of GDP by 2050 without urgent action by one and all.

The average carbon footprint of a person in India is 0.56 tonnes per year, compared to the global average of 4 tonnes. (India's per capita carbon footprint is 60 per cent lower than the global average)

According to the United Nations Environment Programme (UNEP), if **one billion people out of the global population of eight billion adopt environment-friendly behaviours in their daily lives**, global carbon emissions could drop **by approximately 20 per cent.**

Objectives of Mission LiFE:

- **Mission LiFE seeks to translate the vision of LiFE into measurable impact.**
- Mission LiFE is designed with the objective to mobilise at least one billion Indians and **other global citizens to take individual and collective action for protecting and preserving the environment in the period 2022 to 2027.**
- Within India, at least 80% of all villages and urban local bodies are aimed to become environment-friendly by 2028.
- It aims to nudge individuals and communities to practise a lifestyle that is synchronous with nature and does not harm it.
- Those who practice such a lifestyle are recognised as **'Pro Planet People (P3)'**.

As a global programme, Mission LiFE envisions three core shifts in our collective approach towards sustainability. These are:

- **Change in Demand (Phase I):** Nudging individuals across the world to practice simple yet effective environment-friendly actions in their daily lives
- **Change in Supply (Phase II):** Changes in large-scale individual demand are expected to gradually nudge industries and markets to respond and tailor supply and procurement as per the revised demands

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- **Change in Policy (Phase III):** By influencing the demand and supply dynamics of India and the world, the long-term vision of Mission LiFE is to trigger shifts in large-scale industrial and government policies that can support both sustainable consumption and production

The LiFE 21-Day Challenge is launched to enable Indians to take one simple environment-friendly action per day for 21 days and eventually develop an environment-friendly lifestyle.

As a part of Mission LiFE, a comprehensive and non-exhaustive list of **75 individual LiFE actions have been** identified across **7 themes – save water, save energy, reduce waste, reduce e-waste, reduce single-use plastics, adopt sustainable food systems, and adopt healthy lifestyles**

Themes of World Environment Day: 5th June

2024: Our Land, Our Future. We are #GenerationRestoration

2025: Ending Plastic Pollution

Environmental Information, Awareness, Capacity Building and Livelihood Programme (EIACP):

The Environmental Information, Awareness, Capacity Building and Livelihood Programme (EIACP) is one of the Central Sector sub-scheme being implemented in alignment with Mission LiFE. As a part of the current mass mobilization campaign, 60 EIACP Centres are actively engaged in promoting awareness about sustainable actions that individuals can undertake.

Environment Education Programme (EEP): The Environmental Education Programme (EEP) is a **Central Sector sub-scheme** being implemented for imparting non-formal environment education through inter alia initiatives for strengthening Eco-club activities in schools and colleges. Sharing the common goal of promoting sustainable lifestyle, EEP is implemented in full alignment with Mission LiFE, through State/UT-level Implementing Agencies.

MISHTI: “Mangrove Initiative for Shoreline Habitats and Tangible Incomes” was announced in the **Union Budget 2023-24** to promote and conserve mangroves. Mangroves are unique, natural eco-system having very high biological productivity and carbon sequestration potential, besides working as a bio-shield.

Sustainable Development

MISHTI Programme was launched today with active participation of coastal States and UTs.

The Programme will cover approximately 540 sq km area across nine (9) coastal States and four (4) UTs in five years (2023-2028).

It will create around 22.8 million man-days with estimated carbon sink of 4.5 million tons of Carbon. It will also create potential areas for nature tourism and livelihood potential for local communities.

Mangrove plantation drive at more than 75 mangrove sites was also organized.

This drive witnessed the participations through the plantation activities by public representatives, local people, village communities, educational institutes and other stakeholders.

Objective:

To take up mangrove reforestation and afforestation along the coastal districts of India by adopting best practices that already exist in India as well as from other countries including Indonesia.

To develop mangrove associated ecotourism initiatives and livelihood generation in coastal states.

INDIA STATE OF FOREST REPORT 2023:

India State of Forest Report 2023 has been released by Union Minister for **Environment, Forest and Climate Change, Bhupender Yadav**.

- **Released at:** Forest Research Institute, Dehradun
- **Edition:** 18th, released in December 2024
- **Report highlights:**

Area of Total Forest and Tree cover	8,27,357sq km (25.17% of the geographical area of the country) · Forest Cover: 7,15,343sq km (21.76%) · Tree Cover: 1,12,014 sq km (3.41%) Note: · As compared to 2021, there is an increase of 1445 sq km in the total forest and tree cover of the country. · It includes 156 sq km increase in the forest cover and 1289 sq km increase in tree cover.
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Sustainable Development

Maximum increase in forest and tree cover	- Chhattisgarh (684 sq km) - Uttar Pradesh (559 sq km) - Odisha (559 sq km) - Rajasthan (394 sq km)
Maximum increase in forest cover	- Mizoram (242 sq km) - Gujarat (180 sq km) - Odisha (152 sq km)
Largest forest and tree cover (Area wise)	- Madhya Pradesh (85,724 sq km) - Arunachal Pradesh (67,083 sq km) - Maharashtra (65,383 sq km)
Largest forest cover (Area-wise)	- Madhya Pradesh (77,073 sq km) - Arunachal Pradesh (65,882 sq km) - Chhattisgarh (55,812 sq km)
Forest cover with respect to total geographical area	- Lakshadweep (91.33 percent) - Mizoram (85.34 percent) - Andaman & Nicobar Island (81.62 percent)

- The present assessment also reveals that 19 states/UTs **have above 33% of the geographical area under forest cover.**
- Out of these, eight states/UTs namely Mizoram, Lakshadweep, A & N Island, Arunachal Pradesh, Nagaland, Meghalaya, Tripura, and Manipur have forest cover above 75 percent.
- The **total mangrove cover is 4,992 sq km in the country.**